

ITM Skills University – B.Tech

DBMS Assignment 2

Database Keys: Primary, Foreign and Candidate Keys Real-World Receipt Analysis

Student Name	Sasanka Sekhar Kundu
Subject	DBMS Fundamentals
Database	PostgreSQL – dbms_assignment_1

1. Receipt Used for Analysis

For this assignment, a supermarket purchase scenario was analysed. The receipt contains store information, bill information, purchased items, quantities, prices, GST, grand total and payment mode.

```
FRESH MART SUPERMARKET
Kolkata, West Bengal - 700001
GSTIN: 19ABCDE1234F1Z5

Bill No: FM20260823001
Date: 23-08-2026
Time: 01:30 PM
Payment Mode: UPI

Basmati Rice 1kg      Qty: 2      ₹120.00      Total: ₹240.00
Amul Milk 1L          Qty: 3      ₹65.00       Total: ₹195.00
Britannia Biscuits    Qty: 2      ₹30.00       Total: ₹60.00

Subtotal: ₹495.00
GST: ₹24.75
Grand Total: ₹519.75
```

Note: The database data used below is consistent with this receipt scenario.

2. Tables and Attributes Identified

Store: store_id, store_name, address, gstin

Receipt: receipt_id, bill_number, store_id, purchase_date, purchase_time, payment_mode, subtotal, gst, grand_total

Receipt Item: receipt_item_id, receipt_id, item_name, quantity, unit_price, total_price

3. Candidate Keys

Table	Candidate Key	Reason
Store	store_id	System-generated value that uniquely identifies each store.
Store	gstin	A GSTIN is unique to a registered business in this database design.

Receipt	receipt_id	System-generated value that uniquely identifies each receipt row.
Receipt	bill_number	Each transaction is assigned a unique bill number.
Receipt Item	receipt_item_id	System-generated value that uniquely identifies each item row.
Receipt Item	receipt_id + item_name	In this sample design, the combination identifies an item entry within a receipt.

4. Primary Keys and Justification

Table	Primary Key	Justification
Store	store_id	Chosen because it is short, stable, system-generated and uniquely identifies every store.
Receipt	receipt_id	Chosen because it is a simple system-generated identifier and does not depend on business data.
Receipt Item	receipt_item_id	Chosen because it uniquely identifies every individual item record and remains simple to reference.

5. Foreign Keys and Relationships

1. receipt.store_id → store.store_id

This foreign key links each receipt to the store where the transaction took place.

2. receipt_item.receipt_id → receipt.receipt_id

This foreign key links every purchased item to its corresponding receipt.

```

STORE
  store_id (PK)
    |
    v
RECEIPT
  receipt_id (PK)
  store_id (FK)
    |
    v
RECEIPT_ITEM
  receipt_item_id (PK)
  receipt_id (FK)

```

6. PostgreSQL CREATE TABLE Statements

```

CREATE TABLE store (
  store_id SERIAL PRIMARY KEY,
  store_name VARCHAR(100) NOT NULL,
  address TEXT NOT NULL,
  gstin VARCHAR(20) UNIQUE NOT NULL

```

```
);

CREATE TABLE receipt (
    receipt_id SERIAL PRIMARY KEY,
    bill_number VARCHAR(30) UNIQUE NOT NULL,
    store_id INT NOT NULL,
    purchase_date DATE NOT NULL,
    purchase_time TIME NOT NULL,
    payment_mode VARCHAR(30) NOT NULL,
    subtotal NUMERIC(10,2) NOT NULL,
    gst NUMERIC(10,2) NOT NULL,
    grand_total NUMERIC(10,2) NOT NULL,
    CONSTRAINT fk_store
        FOREIGN KEY (store_id)
        REFERENCES store(store_id)
);

CREATE TABLE receipt_item (
    receipt_item_id SERIAL PRIMARY KEY,
    receipt_id INT NOT NULL,
    item_name VARCHAR(100) NOT NULL,
    quantity INT NOT NULL,
    unit_price NUMERIC(10,2) NOT NULL,
    total_price NUMERIC(10,2) NOT NULL,
    CONSTRAINT fk_receipt
        FOREIGN KEY (receipt_id)
        REFERENCES receipt(receipt_id)
);
```

```
dbms_assignment_1=# CREATE TABLE store (
    store_id SERIAL PRIMARY KEY,
    store_name VARCHAR(100) NOT NULL,
    address TEXT NOT NULL,
    gstin VARCHAR(20) UNIQUE NOT NULL
);

CREATE TABLE receipt (
    receipt_id SERIAL PRIMARY KEY,
    bill_number VARCHAR(30) UNIQUE NOT NULL,
    store_id INT NOT NULL,
    purchase_date DATE NOT NULL,
    purchase_time TIME NOT NULL,
    payment_mode VARCHAR(30) NOT NULL,
    subtotal NUMERIC(10,2) NOT NULL,
    gst NUMERIC(10,2) NOT NULL,
    grand_total NUMERIC(10,2) NOT NULL,
    CONSTRAINT fk_store
        FOREIGN KEY (store_id)
        REFERENCES store(store_id)
);

CREATE TABLE receipt_item (
    receipt_item_id SERIAL PRIMARY KEY,
    receipt_id INT NOT NULL,
    item_name VARCHAR(100) NOT NULL,
    quantity INT NOT NULL,
    unit_price NUMERIC(10,2) NOT NULL,
    total_price NUMERIC(10,2) NOT NULL,
    CONSTRAINT fk_receipt
        FOREIGN KEY (receipt_id)
        REFERENCES receipt(receipt_id)
);

CREATE TABLE
CREATE TABLE
CREATE TABLE
dbms_assignment_1=# \dt
      List of tables
Schema | Name      | Type  | Owner
-----|-----|-----|-----
public | receipt   | table | zero
public | receipt_item | table | zero
public | store     | table | zero
(3 rows)

dbms_assignment_1=# INSERT INTO store (store_name, address, gstin)
VALUES (
    'Fresh Mart Supermarket',
    'Kolkata, West Bengal - 700001',
    '19ABCDE1234F125'
);

INSERT 0 1
dbms_assignment_1=# SELECT * FROM store;
 store_id | store_name      | address                | gstin
-----|-----|-----|-----
        1 | Fresh Mart Supermarket | Kolkata, West Bengal - 700001 | 19ABCDE1234F125
(1 row)

dbms_assignment_1=# INSERT INTO receipt (
    bill_number,
    store_id,
    purchase_date,
    purchase_time,
    payment_mode,
    subtotal,
    gst,
    grand_total
)
```

SCREENSHOT 1 – CREATE TABLE SUCCESS

7. PostgreSQL INSERT Statements

```
INSERT INTO store (store_name, address, gstin)
VALUES (
    'Fresh Mart Supermarket',
    'Kolkata, West Bengal - 700001',
    '19ABCDE1234F1Z5'
);

INSERT INTO receipt (
    bill_number, store_id, purchase_date, purchase_time,
    payment_mode, subtotal, gst, grand_total
)
VALUES (
    'FM20260823001', 1, '2026-08-23', '13:30:00',
    'UPI', 495.00, 24.75, 519.75
);

INSERT INTO receipt_item (
    receipt_id, item_name, quantity, unit_price, total_price
)
VALUES
    (1, 'Basmati Rice 1kg', 2, 120.00, 240.00),
    (1, 'Amul Milk 1L', 3, 65.00, 195.00),
    (1, 'Britannia Biscuits', 2, 30.00, 60.00);
```

```
receipt_id INT NOT NULL,
item_name VARCHAR(100) NOT NULL,
quantity INT NOT NULL,
unit_price NUMERIC(10,2) NOT NULL,
total_price NUMERIC(10,2) NOT NULL,

CONSTRAINT fk_receipt
FOREIGN KEY (receipt_id)
REFERENCES receipt(receipt_id)
);

CREATE TABLE
CREATE TABLE
CREATE TABLE
dbms_assignment_1=# \dt
List of tables
Schema | Name | Type | Owner
public | receipt | table | zero
public | receipt_item | table | zero
public | store | table | zero
(3 rows)

dbms_assignment_1=# INSERT INTO store (store_name, address, gstin)
VALUES (
    'Fresh Mart Supermarket',
    'Kolkata, West Bengal - 700001',
    '19ABCDE1234F1Z5'
);
INSERT 0 1
dbms_assignment_1=# SELECT * FROM store;
store_id | store_name | address | gstin
-----
1 | Fresh Mart Supermarket | Kolkata, West Bengal - 700001 | 19ABCDE1234F1Z5
(1 row)

dbms_assignment_1=# INSERT INTO receipt (
    bill_number,
    store_id,
    purchase_date,
    purchase_time,
    payment_mode,
    subtotal,
    gst,
    grand_total
)
VALUES (
    'FM20260823001',
    1,
    '2026-08-23',
    '13:30:00',
    'UPI',
    495.00,
    24.75,
    519.75
);
INSERT 0 1
dbms_assignment_1=# INSERT INTO receipt_item (
    receipt_id,
    item_name,
    quantity,
    unit_price,
    total_price
)
VALUES
    (1, 'Basmati Rice 1kg', 2, 120.00, 240.00),
    (1, 'Amul Milk 1L', 3, 65.00, 195.00),
    (1, 'Britannia Biscuits', 2, 30.00, 60.00);
INSERT 0 3
dbms_assignment_1=# SELECT * FROM store;
```

SCREENSHOT 2 – INSERT SUCCESS

8. Verification Using SELECT Statements

```
SELECT * FROM store;
```

```
SELECT * FROM receipt;
```

```
SELECT * FROM receipt_item;
```

```
dbms_assignment_1=# INSERT INTO store (store_name, address, gstin)
VALUES (
    'Fresh Mart Supermarket',
    'Kolkata, West Bengal - 700001',
    '19ABCDE1234F125'
);
INSERT 0 1
dbms_assignment_1=# SELECT * FROM store;
 store_id | store_name | address | gstin
-----|-----|-----|-----
      1 | Fresh Mart Supermarket | Kolkata, West Bengal - 700001 | 19ABCDE1234F125
(1 row)

dbms_assignment_1=# INSERT INTO receipt (
    bill_number,
    store_id,
    purchase_date,
    purchase_time,
    payment_mode,
    subtotal,
    gst,
    grand_total
)
VALUES (
    'FM20260823001',
    1,
    '2026-08-23',
    '13:30:00',
    'UPI',
    495.00,
    24.75,
    519.75
);
INSERT 0 1
dbms_assignment_1=# INSERT INTO receipt_item (
    receipt_id,
    item_name,
    quantity,
    unit_price,
    total_price
)
VALUES
    (1, 'Basmati Rice 1kg', 2, 120.00, 240.00),
    (1, 'Amul Milk 1L', 3, 65.00, 195.00),
    (1, 'Britannia Biscuits', 2, 30.00, 60.00);
INSERT 0 3
dbms_assignment_1=# SELECT * FROM store;

SELECT * FROM receipt;

SELECT * FROM receipt_item;
 receipt_id | store_id | address | gstin
-----|-----|-----|-----
      1 | Fresh Mart Supermarket | Kolkata, West Bengal - 700001 | 19ABCDE1234F125
(1 row)

 receipt_id | bill_number | store_id | purchase_date | purchase_time | payment_mode | subtotal | gst | grand_total
-----|-----|-----|-----|-----|-----|-----|-----|-----
      1 | FM20260823001 | 1 | 2026-08-23 | 13:30:00 | UPI | 495.00 | 24.75 | 519.75
(1 row)

 receipt_item_id | receipt_id | item_name | quantity | unit_price | total_price
-----|-----|-----|-----|-----|-----
      1 | 1 | Basmati Rice 1kg | 2 | 120.00 | 240.00
      2 | 1 | Amul Milk 1L | 3 | 65.00 | 195.00
      3 | 1 | Britannia Biscuits | 2 | 30.00 | 60.00
(3 rows)

dbms_assignment_1=#
```

SCREENSHOT 3 – SELECT RESULTS

9. Summary

This assignment analyses a supermarket receipt and converts its information into a relational database design. The receipt data was divided into three logical tables: Store, Receipt and Receipt_Item. The Store table keeps information about the supermarket, while the Receipt table stores details of the complete transaction. The Receipt_Item table stores every individual product purchased in that transaction. For the Store table, both store_id and GSTIN can uniquely identify a store, so they are candidate keys. store_id was selected as the primary key because it is short, system-generated and stable. In the Receipt table, receipt_id and bill_number are candidate keys. receipt_id was selected as the primary key for the same reason. In the Receipt_Item table, receipt_item_id uniquely identifies each item row and was selected as the primary key. The store_id column in Receipt is a foreign key referencing Store, which links every bill to its store. Similarly, receipt_id in Receipt_Item is a foreign key referencing Receipt, linking each purchased item to the correct transaction. These keys reduce duplication and maintain proper relationships and data integrity in the PostgreSQL database.

10. Conclusion

The PostgreSQL statements were successfully executed in the dbms_assignment_1 database. The three tables were created, the receipt data was inserted, and the stored records were verified using SELECT queries.